

Introduction

This Technical Design Document outlines the architectural, functional, and data design specifications for two web-based systems: the Online Store and the Loaning Website. These systems aim to provide responsive, secure and user-friendly platforms that address distinct business needs online product sales and loan application management, while sharing a modern, scalable technology stack. The document details each project's requirements, system architecture, entity relationships and process flows, accompanied by visual diagrams to facilitate clear understanding among developers, stakeholders and future maintainers. Technologies such as React.js, Firebase, Supabase (PostgreSQL) and integration with external services like WhatsApp and email notifications have been chosen to ensure reliability, performance and ease of maintenance. By presenting both projects individually within a consistent framework, this document serves as a blueprint for development, testing, deployment and long-term support.

Online Store – Technical Design

1. Overview

Purpose: Develop a responsive online platform where users can browse a product catalog and place orders via WhatsApp.

Stakeholders: Business owner, customers, admin staff.

Core Features:

- Product catalog with descriptions & prices
- User registration & login (Supabase Auth)
- WhatsApp redirect for order placement
- Admin panel to manage products

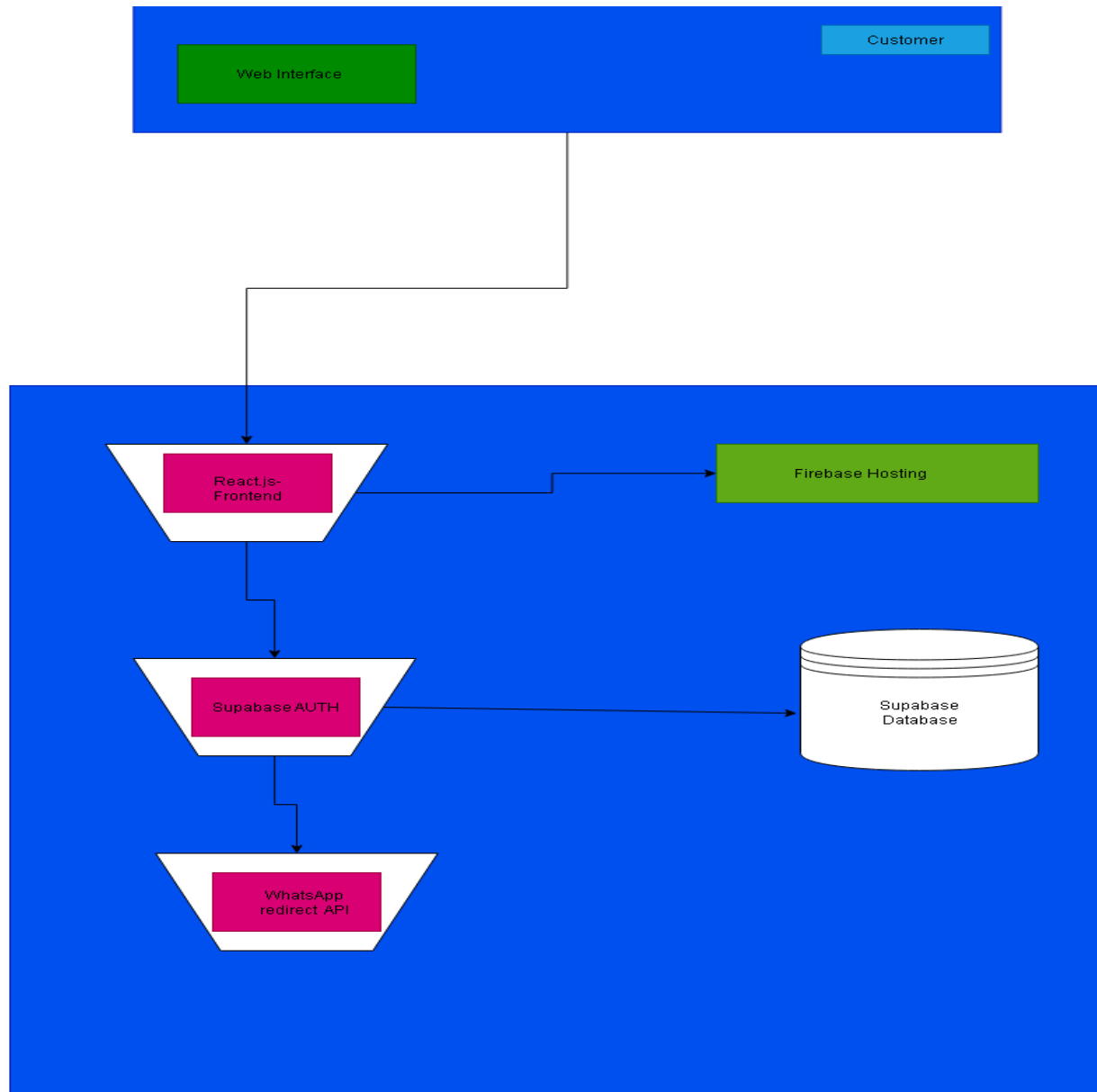
2. System Architecture

Technology Stack:

- Frontend: HTML, CSS, JavaScript, React.js
- Backend: Firebase (hosting & serverless functions)
- Database: Supabase (PostgreSQL)

- Auth: Supabase Auth
- Payments: WhatsApp redirect link
- Design Tool: Figma

Architecture Diagram:

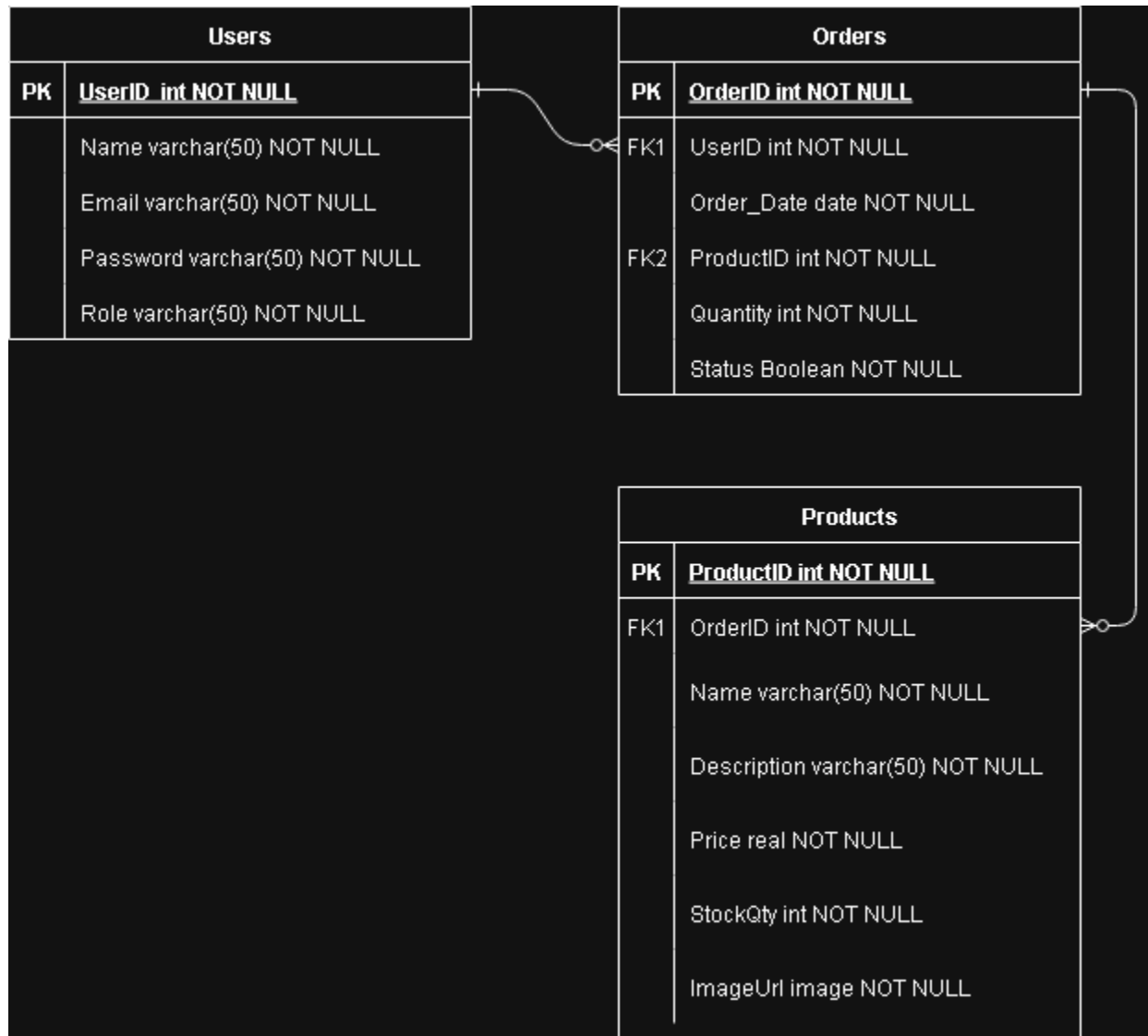


3. Data Design

Entities:

- Users (UserID, Name, Email, PasswordHash, Role)
- Products (ProductID, Name, Description, Price, StockQty, ImageURL)
- Orders (OrderID, UserID, ProductID, Quantity, OrderDate, Status)

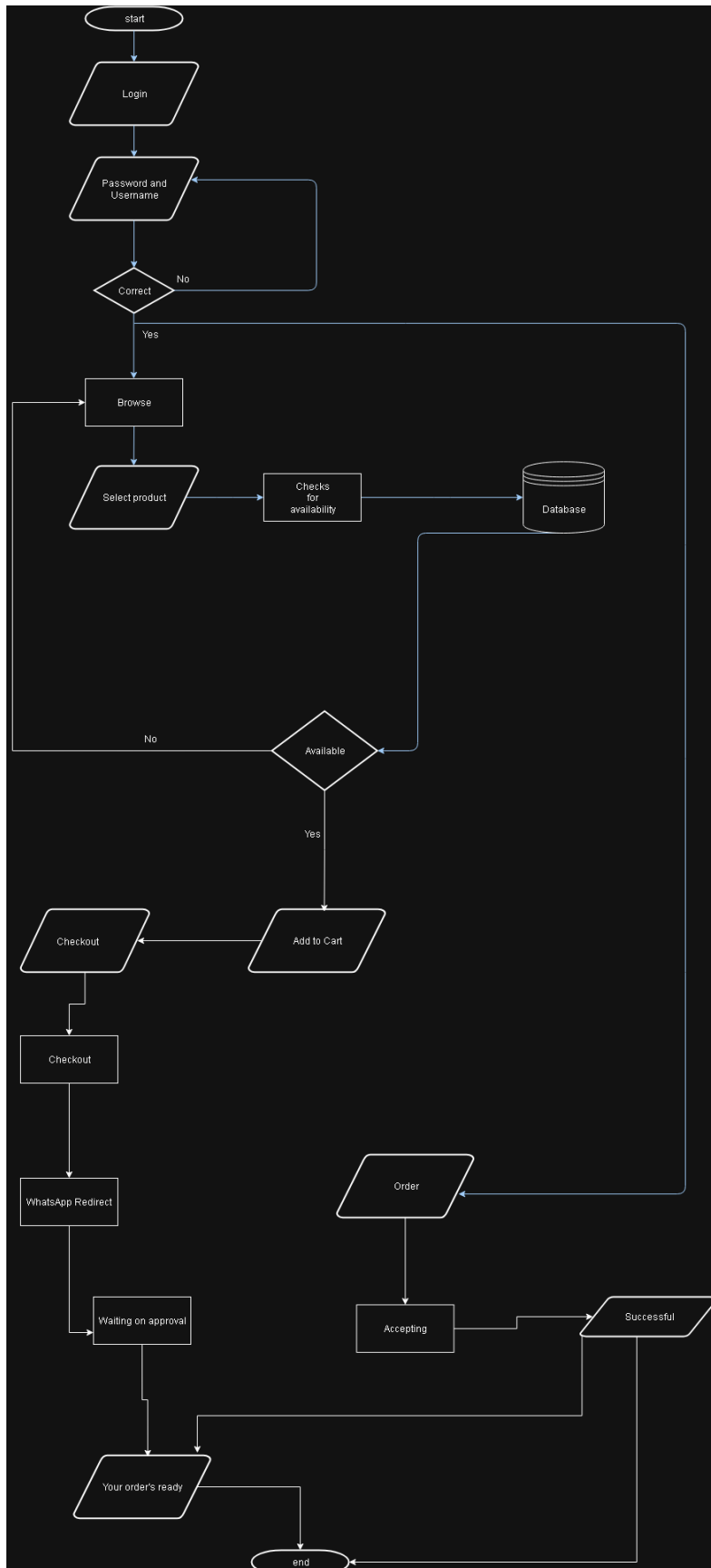
ERD:



4. Sequence Example – Order Process

1. User logs in.
2. User browses catalog.
3. User clicks “Order via WhatsApp”.
4. System redirects to WhatsApp with pre-filled order message.

5. Admin receives message and processes manually.



5. Security

- Passwords hashed via Supabase Auth
- HTTPS enforced via Firebase Hosting

Loaning Website – Technical Design

1. Overview

Purpose: Allow customers to apply for personal loans online and track application progress.

Stakeholders: Loan applicants, admin staff.

Core Features:

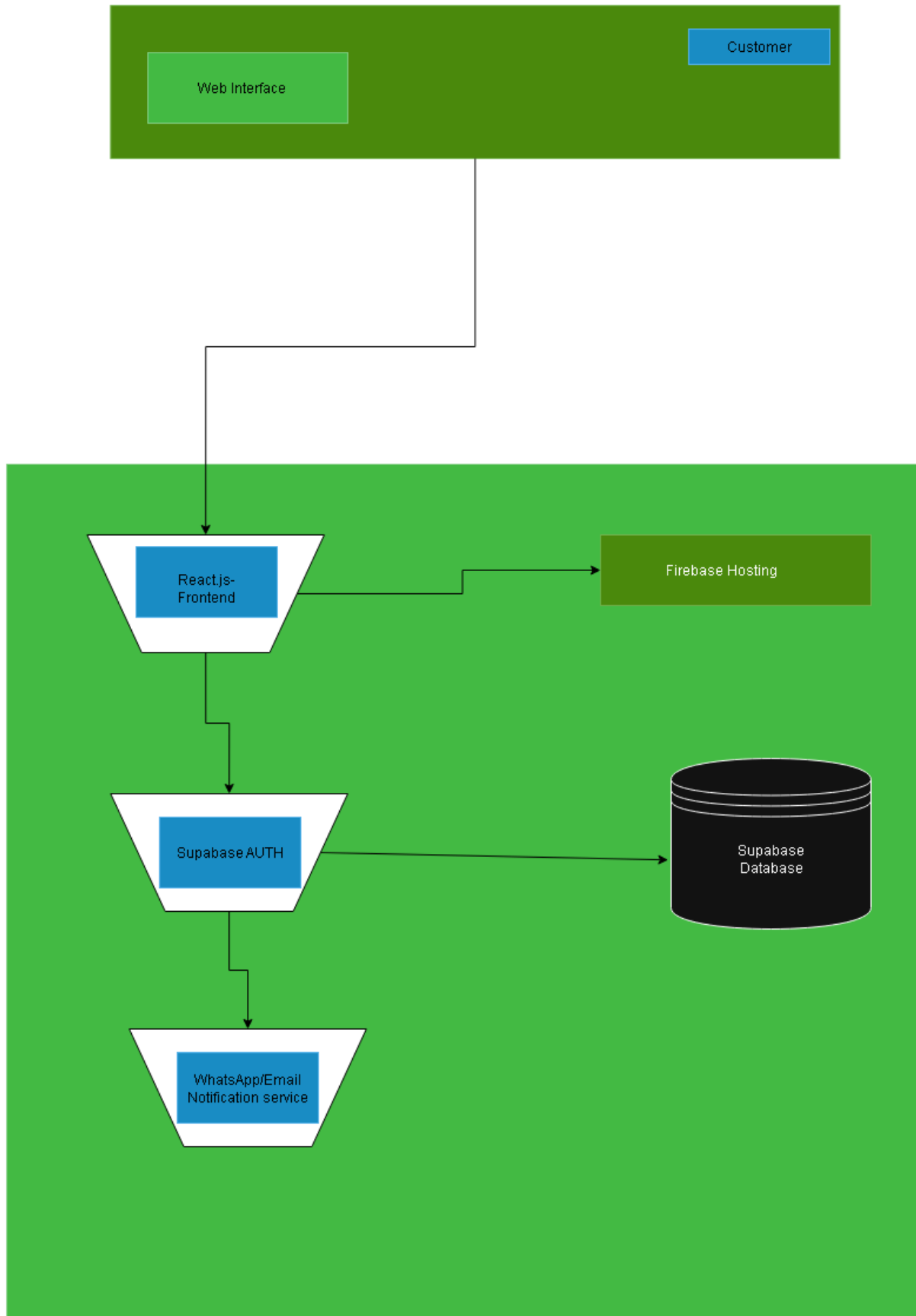
- Loan application form
- Admin dashboard for reviewing applications
- Authentication via Supabase Auth
- Document upload support
- Notifications via WhatsApp or email

2. System Architecture

Technology Stack:

- Frontend: HTML, CSS, JavaScript, React.js
- Backend: Firebase
- Database: Supabase (PostgreSQL)
- Auth: Supabase Auth
- Notifications: WhatsApp redirect / Email
- Design Tool: Figma

Architecture Diagram:

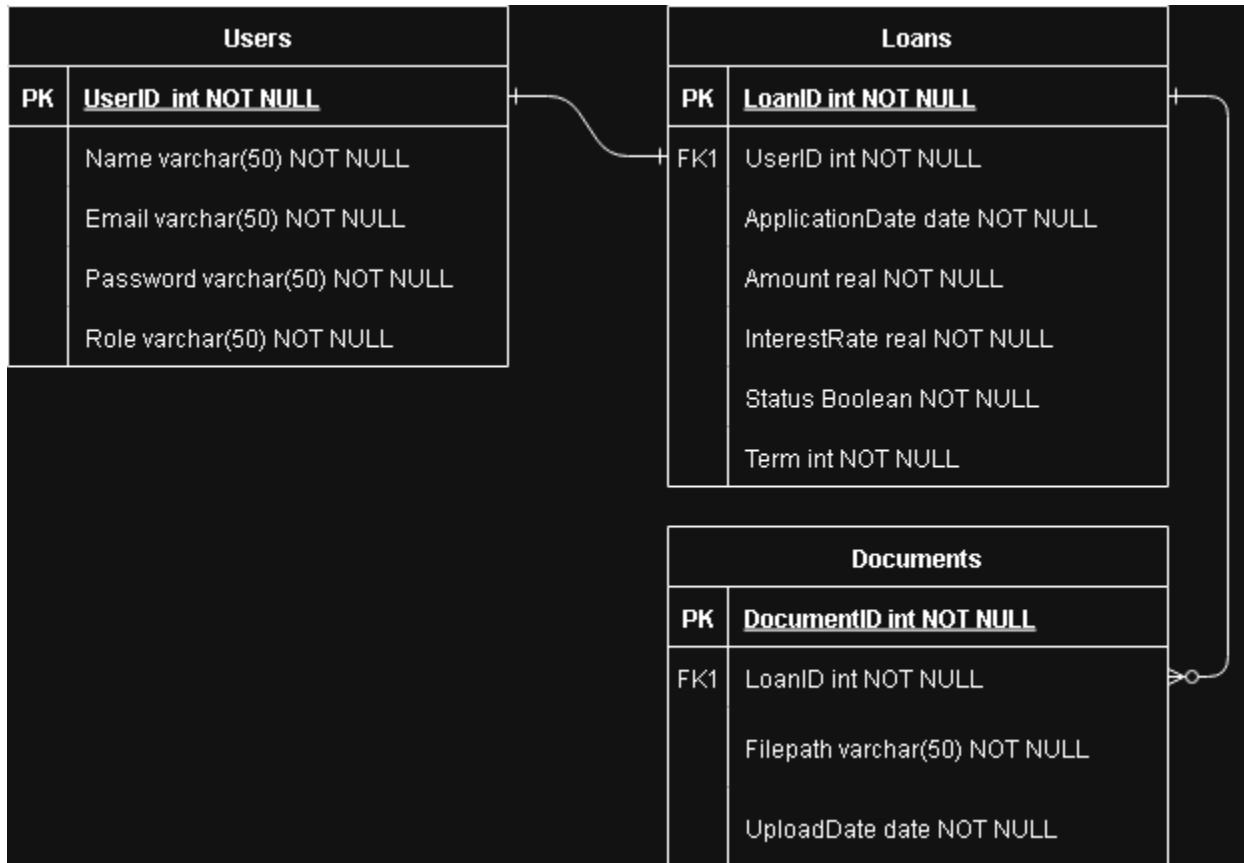


3. Data Design

Entities:

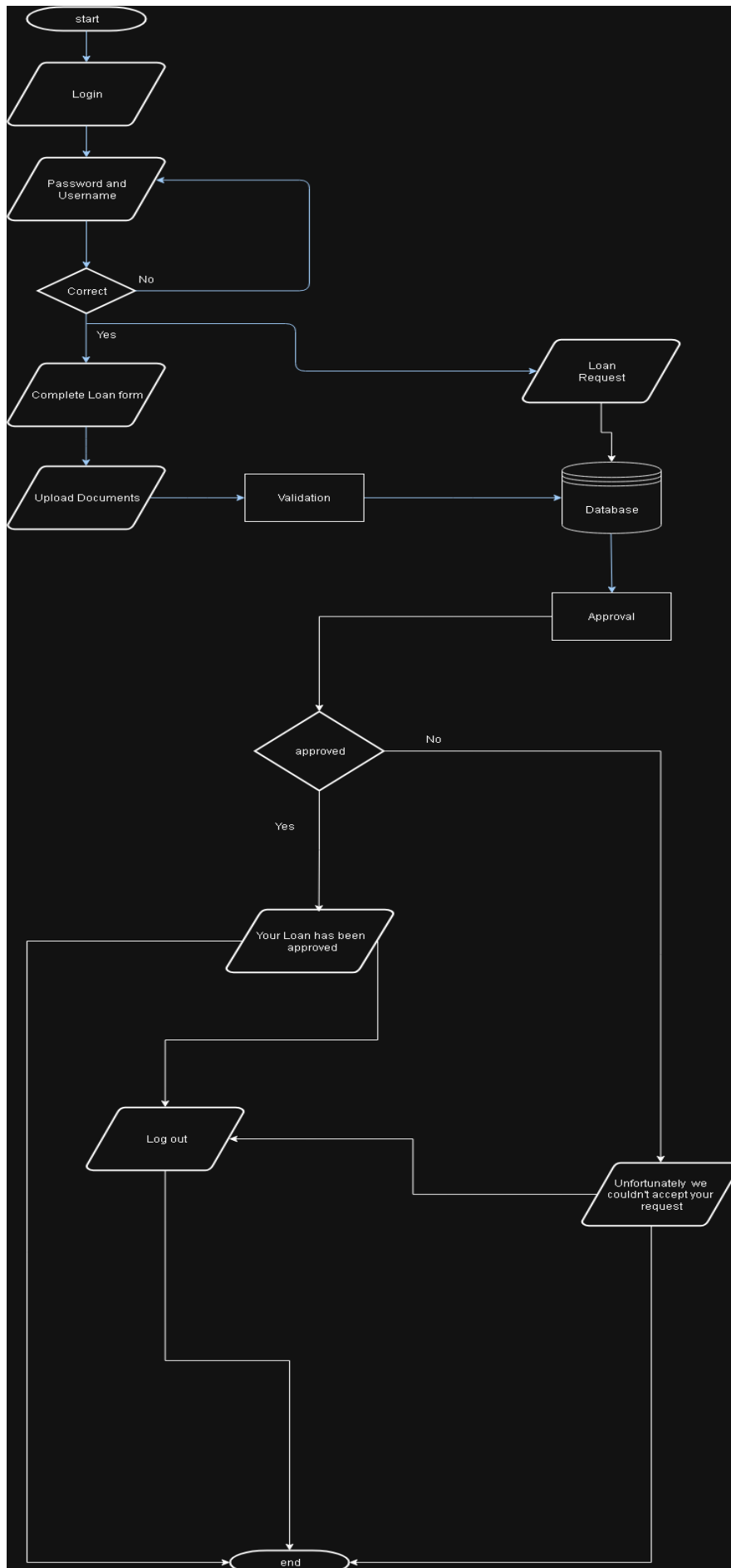
- Users (UserID, Name, Email, Phone, PasswordHash, Role)
- Loans (LoanID, UserID, Amount, Term, InterestRate, Status, ApplicationDate)
- Documents (DocumentID, LoanID, FilePath, UploadedDate)

ERD:



4. Sequence Example – Loan Application Process

1. User logs in.
2. User completes loan form & uploads documents.
3. System stores application in Supabase DB.
4. Admin reviews application via dashboard.
5. Admin approves/rejects and system sends notification.



5. Security

- Supabase Auth for secure login
- Document files stored with restricted access
- Role-based access control for admin vs. users

Conclusion

The technical designs presented in this document provide a clear roadmap for the successful development and deployment of the Online Store and Loaning Website. Each system's architecture, database design, and process flows have been carefully planned to align with the business objectives, ensure data integrity, and deliver an intuitive user experience. Leveraging the chosen technology stack, these solutions are positioned to scale, adapt, and integrate with emerging tools and services. The combination of detailed specifications and visual diagrams will guide the development team, reduce implementation risks, and support ongoing enhancements. With this foundation, both projects are well-prepared to progress into the implementation and testing phases with clarity and confidence.